Ungaugeability of Kosmann Symmetries of the Dirac Operator on Compact Lie Groups

MemeTheory Claude Opus 4.6

March 20, 2026

Abstract

We prove that the $U(1)$ symmetry generated by the Kosmann derivative K_7 of the Dirac operator on Jensen-deformed SU(3) cannot be gauged within Connes' noncommutative geometry framework. Working with the almost-commutative spectral triple $\mathcal{A} = C^\infty(\mathrm{SU}(3)) \otimes \mathcal{A}_F$ with $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ (the Standard Model algebra), we show that K_7 is excluded from the gauge sector by two structurally distinct arguments. First, the commutant obstruction: $[K_7, D_K(\tau)] = 0$ for all Jensen parameter values τ, so K_7 generates a trivial one-form in $\Omega^1_D(\mathcal{A})$. By the spectral theorem, this extends to all functionals of D_K, closing the obstruction non-perturbatively. Second, a categorical argument independent of the commutant property: K_7 generates a diffeomorphism of SU(3) and therefore belongs to $\mathrm{Aut}(C^\infty(\mathrm{SU}(3)))$; since $C^\infty(\mathrm{SU}(3))$ is commutative, $\mathrm{Inn}(C^\infty(\mathrm{SU}(3))) = \{\mathrm{id}\}$ and all its automorphisms are outer; gauge fields arise from $\mathrm{Inn}(\mathcal{A}_F)$, not from diffeomorphisms of the base. The inner/outer distinction is classical in Connes' framework; what is new is an explicit worked example demonstrating ungaugeability of a Kosmann symmetry in a concrete Kaluza–Klein setting, with computational verification. The Anderson–Higgs impossibility for $U(1)_7$ follows as a direct corollary.

1 Introduction

A central achievement of Connes' noncommutative geometry (NCG) is the derivation of gauge theories from purely geometric data Connes [1996], Chamseddine and Connes [1996]. Given a real spectral triple $(\mathcal{A}, \mathcal{H}, D, J, \gamma)$, gauge fields arise as *inner fluctuations* of the Dirac operator:

$$D \longmapsto D_A = D + A + \varepsilon' JAJ^{-1}, \qquad A = \sum_i a_i[D, b_i], \quad a_i, b_i \in \mathcal{A}. \tag{1}$$

The self-adjoint one-form A belongs to the space $\Omega^1_D(\mathcal{A})$ of *universal one-forms* evaluated through the representation of D. The gauge group is the quotient $\mathcal{G} = U(\mathcal{A})/U(Z(\mathcal{A}))$, where $U(\mathcal{A})$ denotes the unitary group of $\mathcal{A}$ and $Z(\mathcal{A})$ its center Connes [1996]. In the almost-commutative geometry $M^4 \times F$ that recovers the Standard Model, the gauge group

emerges as $U(1)_Y \times \mathrm{SU}(2)_L \times \mathrm{SU}(3)_c$, and the Higgs field appears as the component of the inner fluctuation along the finite direction F Chamseddine et al. [2007].

This framework draws a sharp line between two classes of symmetries. *Inner automorphisms* of $\mathcal{A}$ – conjugation by unitaries $u \in U(\mathcal{A})$ – generate gauge transformations and produce nontrivial gauge fields through the map $b \mapsto [D, b]$. *Outer automorphisms* – elements of $\mathrm{Aut}(\mathcal{A})/\mathrm{Inn}(\mathcal{A})$ – correspond to diffeomorphisms of the underlying space and do *not* generate gauge fields. This inner/outer dichotomy, established in Connes' foundational work Connes [1994, 1996], is the noncommutative-geometric realization of the physical distinction between gauge symmetries and spacetime symmetries.

The question we address is: *can a symmetry of the Dirac operator that arises from a diffeomorphism of the underlying manifold ever be promoted to a gauge symmetry through any mechanism available within the spectral action framework?*

We answer this question in the negative for a specific, concrete case. Consider the Dirac operator D_K on SU(3) equipped with a one-parameter family of left-invariant metrics – the Jensen deformation Jensen [1973], parametrized by $\tau \geq 0$. At $\tau = 0$, the metric is bi-invariant and the full $\mathrm{SU}(3) \times \mathrm{SU}(3)$ isometry group acts on the spinor bundle through Kosmann derivatives Kosmann [1971]. For $\tau > 0$, the Jensen deformation breaks $\mathrm{SU}(3) \to U(1)_7$, where $U(1)_7$ is the subgroup generated by the seventh Gell-Mann matrix λ_7. The corresponding Kosmann derivative K_7 continues to commute with $D_K(\tau)$ at every τ:

$$[iK_7,\, D_K(\tau)] = 0, \qquad \forall\, \tau \geq 0. \tag{2}$$

This identity has been verified numerically to machine precision ($< 10^{-13}$) across 79,968 matrix element pairs and at 9 values of τ in $[0, 0.50]$ Project computation [2026c].

Despite being an exact symmetry of the Dirac operator, $U(1)_7$ cannot be gauged. We establish this through two structurally distinct arguments, each sufficient on its own:

1. **Commutant obstruction**: K_7 lies in the commutant of D_K, hence $a[D_K, K_7] = 0$ for all a in any algebra $\mathcal{A}$. The one-form $\Omega^1_D(\mathcal{A})$ in the K_7 direction is trivial. By the spectral theorem, this extends non-perturbatively to all functionals of D_K, including quantum corrections at every loop order.

2. **Categorical**: K_7 generates a diffeomorphism of SU(3), which is an outer automorphism of $C^\infty(\mathrm{SU}(3))$ (since all automorphisms of a commutative algebra are outer). In the almost-commutative spectral triple $\mathcal{A} = C^\infty(\mathrm{SU}(3)) \otimes \mathcal{A}_F$, gauge fields arise from $\mathrm{Inn}(\mathcal{A}_F)$, not from diffeomorphisms of the base. This argument does *not* use $[K_7, D_K] = 0$.

The physical consequence is that $U(1)_7$ admits no Anderson–Higgs mechanism: there is no gauge boson to absorb the Goldstone mode of $U(1)_7$ breaking, no mass generation through the standard Higgs mechanism, and no gauge kinetic term $F_7^{\mu\nu} F_{7,\mu\nu}$ in the spectral action.

The distinction between inner and outer automorphisms in NCG is well established Connes [1994, 1996], Connes and Marcolli [2008]. For the commutative algebra $C^\infty(\mathrm{SU}(3))$ alone, ungaugeability of diffeomorphisms follows from general principles. What is new in this work is an explicit worked example demonstrating ungaugeability of a Kosmann symmetry in

a concrete Kaluza–Klein setting – the Jensen-deformed SU(3) with an almost-commutative spectral triple – verified computationally and established by two structurally independent arguments.

2 Preliminaries

2.1 Real spectral triples

We recall the essential definitions following Connes Connes [1995, 1996].

Definition 1 (Real spectral triple)**.** A *real spectral triple* of KO-dimension $n \bmod 8$ is a quintuple $(\mathcal{A}, \mathcal{H}, D, J, \gamma)$ where:

- $\mathcal{A}$ is a unital $*$-algebra represented faithfully on a separable Hilbert space $\mathcal{H}$;
- D is a self-adjoint operator on $\mathcal{H}$ with compact resolvent, such that $[D, a]$ is bounded for all $a \in \mathcal{A}$;
- $J : \mathcal{H} \to \mathcal{H}$ is an antiunitary operator (the *real structure*);
- γ is a self-adjoint unitary ($\mathbb{Z}/2$-grading) when n is even;

subject to the sign conditions

$$J^2 = \varepsilon, \quad JD = \varepsilon' DJ, \quad J\gamma = \varepsilon'' \gamma J, \tag{3}$$

where $(\varepsilon, \varepsilon', \varepsilon'') \in \{+1, -1\}^3$ are determined by $n \bmod 8$ according to the KO-dimension table Connes [1995].

The KO-dimension relevant to the Standard Model is $n = 6 \bmod 8$, corresponding to $(\varepsilon, \varepsilon', \varepsilon'') = (+1, +1, -1)$.

2.2 The almost-commutative spectral triple on SU(3)

The paper works throughout with an *almost-commutative* spectral triple, which we now specify. The purely commutative algebra $C^\infty(\mathrm{SU}(3))$ has a trivial inner automorphism group: for any commutative algebra $\mathcal{A}$, every automorphism $\alpha(a) = uau^* = a$ is the identity when $\mathcal{A}$ is commutative, so $\mathrm{Inn}(\mathcal{A}) = \{\mathrm{id}\}$. Consequently, the gauge group of a purely commutative spectral triple is trivial and no symmetry can be gauged – making any ungaugeability result vacuous. To obtain a nontrivial gauge sector, we work with the product geometry:

Definition 2 (Almost-commutative spectral triple on SU(3))**.** The spectral triple is $(\mathcal{A}, \mathcal{H}, D, J, \gamma)$ where:

- The algebra is $\mathcal{A} = C^\infty(\mathrm{SU}(3)) \otimes \mathcal{A}_F$ with finite algebra $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$;
- The Hilbert space is $\mathcal{H} = L^2(\mathrm{SU}(3), \Sigma) \otimes \mathcal{H}_F$ where $\mathcal{H}_F = \mathbb{C}^{32}$ is the Standard Model fermion space (one generation, particle and antiparticle);

- The Dirac operator is $D = D_K(\tau) \otimes \mathbf{1}_F + \gamma_9 \otimes D_F$, where $D_K(\tau)$ is the Dirac operator on $(\mathrm{SU}(3), g_\tau)$, γ_9 is the chirality operator on SU(3), and D_F is the finite Dirac operator encoding fermion masses and mixings;
- $J = J_{\mathrm{SU}(3)} \otimes J_F$ and $\gamma = \gamma_9 \otimes \gamma_F$ are the product real structure and grading.

The gauge group of this spectral triple is

$$\mathcal{G} = \mathrm{Inn}(\mathcal{A}) = \mathrm{Inn}(\mathcal{A}_F) \cong U(1)_Y \times \mathrm{SU}(2)_L \times \mathrm{SU}(3)_c, \tag{4}$$

where the equality $\mathrm{Inn}(\mathcal{A}) = \mathrm{Inn}(\mathcal{A}_F)$ follows from the commutativity of $C^\infty(\mathrm{SU}(3))$: inner automorphisms can only arise from the noncommutative factor $\mathcal{A}_F$. The Standard Model gauge fields – the hypercharge, weak, and color gauge bosons – emerge as inner fluctuations of D from this algebra.

The Kosmann derivative K_7 acts on the *first* tensor factor: it is an automorphism of $C^\infty(\mathrm{SU}(3))$, not of $\mathcal{A}_F$. Since $C^\infty(\mathrm{SU}(3))$ is commutative, $K_7 \in \mathrm{Aut}(C^\infty(\mathrm{SU}(3)))$ is outer – in fact, *every* nontrivial automorphism of a commutative algebra is outer. Even in the product $\mathcal{A} = C^\infty(\mathrm{SU}(3)) \otimes \mathcal{A}_F$, the element $K_7 \otimes \mathbf{1}_F$ acts trivially on $\mathcal{A}_F$ and as an outer automorphism on $C^\infty(\mathrm{SU}(3))$, so it remains outer for the full algebra. This makes the ungaugeability result nontrivial: the SM gauge symmetries CAN be gauged (they come from $\mathrm{Inn}(\mathcal{A}_F)$), but $U(1)_7$ CANNOT (it comes from a diffeomorphism of SU(3)).

2.3 Inner automorphisms and gauge fields

The automorphism group $\mathrm{Aut}(\mathcal{A})$ of the algebra acts on the spectral triple by transforming $D \mapsto \alpha(D)$ for $\alpha \in \mathrm{Aut}(\mathcal{A})$. The subgroup of *inner automorphisms* $\mathrm{Inn}(\mathcal{A})$ consists of conjugations $a \mapsto uau^*$ by unitaries $u \in U(\mathcal{A})$.

In Connes' framework Connes [1996, 1994], the gauge group of the spectral triple is

$$\mathcal{G} = \mathrm{Inn}(\mathcal{A}) \cong U(\mathcal{A})/U(Z(\mathcal{A})), \tag{5}$$

where $Z(\mathcal{A})$ denotes the center of $\mathcal{A}$. Gauge connections are inner fluctuations of the Dirac operator (1), and the space of gauge potentials is the space of self-adjoint one-forms

$$\Omega^1_D(\mathcal{A}) = \Big\{ \sum_i a_i[D, b_i] : a_i, b_i \in \mathcal{A} \Big\}. \tag{6}$$

The fundamental principle is that an element $b \in \mathcal{A}$ generates a nontrivial gauge field if and only if $[D, b] \neq 0$. Elements in the commutant $\{b \in \mathcal{A} : [D, b] = 0\}$ generate *zero* gauge fields.

Remark 3 (Inner vs. outer). Outer automorphisms – elements of $\mathrm{Out}(\mathcal{A}) = \mathrm{Aut}(\mathcal{A})/\mathrm{Inn}(\mathcal{A})$ – correspond to diffeomorphisms of the underlying space in the commutative case ($\mathcal{A} = C^\infty(M)$, where $\mathrm{Aut}(\mathcal{A}) \cong \mathrm{Diff}(M)$). These do *not* generate gauge fields. The full automorphism group of a spectral triple decomposes as

$$\mathrm{Aut}(\mathcal{A}) = \mathrm{Inn}(\mathcal{A}) \rtimes \mathrm{Out}(\mathcal{A}) \longleftrightarrow \text{gauge} \times \text{diffeos}.$$

This is the noncommutative-geometric formulation of the equivalence principle: gravity (diffeomorphisms) and gauge forces (inner automorphisms) are geometrically unified but categorically distinct.

2.4 The spectral action

The bosonic action functional is the spectral action Chamseddine and Connes [1996]:

$$S_b = \mathrm{Tr}\ f\left(\frac{D_A^2}{\Lambda^2}\right), \tag{7}$$

where $f : \mathbb{R}_+ \to \mathbb{R}_+$ is a smooth, positive, decreasing cutoff function and Λ is an energy scale. The asymptotic expansion for $\Lambda \to \infty$ yields

$$S_b \sim 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4 + O(\Lambda^{-2}), \tag{8}$$

where $f_{2k} = \int_0^\infty f(u)\, u^{k-1}\, du$ are the moments of f, and a_{2k} are the Seeley–DeWitt coefficients of D_A^2. For the almost-commutative Standard Model geometry, a_0 gives the cosmological constant, a_2 gives the Einstein–Hilbert action with a Higgs mass term, and a_4 gives the Yang–Mills, Higgs quartic, and gravitational curvature-squared terms Chamseddine et al. [2007].

A gauge kinetic term $\frac{1}{g_a^2}\int |F_a|^2\, d\mathrm{vol}$ for a gauge field A_a appears in a_4 if and only if the corresponding generator produces a nontrivial one-form through (6). If $[D, b] = 0$ for the generator b, no such term arises.

2.5 The Kosmann derivative

Let (M, g) be a Riemannian manifold with spin structure, and let X be a Killing vector field. The *Kosmann derivative* Kosmann [1971] is the canonical lift of X to an operator on the spinor bundle ΣM:

$$K_X = \nabla_X^\Sigma + \tfrac{1}{4}\,(\nabla_a X_b)\,\gamma^a\gamma^b, \tag{9}$$

where ∇^Σ is the spin connection and γ^a are the Clifford generators in an orthonormal frame. When X is a Killing field, the Killing equation $\nabla_a X_b + \nabla_b X_a = 0$ guarantees that the second term is antisymmetric in a, b and thus involves only the $\gamma^{[a}\gamma^{b]}$ component of the Clifford product. The Kosmann derivative satisfies the Leibniz rule with respect to Clifford multiplication and preserves the spin structure. For a Killing field, K_X commutes with the Dirac operator:

$$[K_X, D] = 0 \qquad \text{whenever } X \text{ is Killing.} \tag{10}$$

This is a classical result Kosmann [1971], Lichnerowicz [1963].

The Kosmann derivative generates an infinitesimal isometry of the spin Riemannian manifold. It is a first-order differential operator on sections of ΣM, encoding the action of the isometry group on spinors. In the language of NCG, the Kosmann derivative is a derivation of the algebra $C^\infty(M)$ that preserves the Dirac operator – but it is an *outer* derivation, not an inner one, because diffeomorphisms of M are outer automorphisms of $C^\infty(M)$.

2.6 Jensen deformation of SU(3)

The Jensen deformation Jensen [1973] is a one-parameter family of left-invariant Riemannian metrics on a compact Lie group G, defined as follows. Fix a closed subgroup $H \subset G$ and

let $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$ be the reductive decomposition of the Lie algebra. The Jensen metric g_τ is defined on $\mathfrak{g}$ by

$$g_\tau(X,Y) = \begin{cases} e^{-4\tau}\, B(X,Y) & \text{if } X,Y \in \mathfrak{h}, \\ e^{\tau}\, B(X,Y) & \text{if } X,Y \in \mathfrak{m}, \\ 0 & \text{if } X \in \mathfrak{h},\ Y \in \mathfrak{m}, \end{cases} \tag{11}$$

where B is the Killing form (normalized to be positive definite) and $\tau \geq 0$ is the deformation parameter. At $\tau = 0$, the metric is proportional to the bi-invariant Killing metric. For $\tau > 0$, the metric remains left-invariant but breaks right-invariance.

For $G = \mathrm{SU}(3)$ and $H = U(1)$ generated by the seventh Gell-Mann matrix λ_7, the Jensen deformation breaks the isometry group as

$$\mathrm{SU}(3) \times \mathrm{SU}(3) \xrightarrow{\tau>0} \mathrm{SU}(3)_L \times U(1)_7, \tag{12}$$

where $\mathrm{SU}(3)_L$ is the left-multiplication isometry (preserved for any left-invariant metric) and $U(1)_7$ is the residual right-multiplication isometry generated by λ_7. The deformation is volume-preserving: $\det(g_\tau)/\det(g_0) = 1$ for all τ Project computation [2026b].

3 The $U(1)_7$ Symmetry of D_K

Let $D_K(\tau)$ denote the Dirac operator on $(\mathrm{SU}(3), g_\tau)$ for the Jensen metric (11). The Kosmann derivative K_7 associated to the Killing field λ_7 satisfies (10) at $\tau = 0$ by the general theory. The nontrivial content is that this commutation relation survives the Jensen deformation.

Proposition 4. *For all $\tau \geq 0$,*

$$[iK_7,\, D_K(\tau)] = 0. \tag{13}$$

Proof. The Killing field $X_7 = \lambda_7$ generates a one-parameter subgroup that remains an isometry of g_τ for all τ, since λ_7 generates the subgroup $H = U(1)$ that defines the Jensen deformation (11). The decomposition $\mathfrak{su}(3) = \mathfrak{h} \oplus \mathfrak{m}$ is $\mathrm{ad}(\lambda_7)$-invariant by construction. The Jensen metric g_τ rescales $\mathfrak{h}$ and $\mathfrak{m}$ by different factors, but this rescaling commutes with $\mathrm{ad}(\lambda_7)$ because $\lambda_7 \in \mathfrak{h}$. Therefore X_7 remains Killing for g_τ at every τ, and the general identity (10) applies.

Numerically, the identity has been verified on the 16-dimensional singlet sector of the Peter–Weyl decomposition of $L^2(\mathrm{SU}(3), \Sigma)$ at 9 values of $\tau \in [0, 0.50]$, with

$$\frac{\|[iK_7, D_K(\tau)]\|_F}{\|D_K(\tau)\|_F} < 3.3 \times 10^{-13}$$

across all 79,968 matrix element pairs Project computation [2026c]. □

Remark 5 (Charge decomposition). The operator iK_7 has eigenvalues $q \in \{0, \pm\frac{1}{4}\}$ on the singlet sector. This induces a charge decomposition of the Dirac eigenspace into three branches:

- B_1: charge $q = 0$ (four modes, $U(2)$ singlet),
- B_2: charge $q = \pm\frac{1}{4}$ (four modes, $U(2)$ doublet),

- B_3: charge $q = 0$ (eight modes, $U(2)$ adjoint).

The charge eigenvalues remain exactly $\pm 1/4$ to machine precision ($< 3 \times 10^{-16}$) at all τ Project computation [2026d]. All spectral selection rules – including the vanishing of the pairing kernel $V(B_1, B_1) = 0$ (the "Trap 1" selection rule) – follow from $U(1)_7$ charge conservation Project computation [2026c].

4 Two Proofs of Ungaugeability

We establish the main result through two structurally distinct arguments. The first exploits the commutant property $[K_7, D_K] = 0$; the second is independent of it.

Theorem 6 (Ungaugeability of $U(1)_7$)**.** *Let $(\mathcal{A}, \mathcal{H}, D, J, \gamma)$ be the almost-commutative spectral triple of Definition 2 on* SU(3) *with Jensen metric g_τ. Then the $U(1)_7$ symmetry generated by K_7 cannot be gauged: there exists no inner fluctuation $A \in \Omega^1_D(\mathcal{A})$ that produces a nontrivial gauge field for $U(1)_7$. Consequently, $U(1)_7$ admits no Anderson–Higgs mechanism within this spectral triple.*

4.1 Proof I: Commutant obstruction

Proof I. By Proposition 4, K_7 lies in the commutant of $D_K(\tau)$:

$$K_7 \in \{T \in \mathcal{B}(\mathcal{H}) : [D_K, T] = 0\}.$$

The inner fluctuation in the K_7 direction, for any $a \in \mathcal{A}$, is

$$A_7 = a \cdot [D_K, K_7] = a \cdot 0 = 0. \tag{14}$$

Therefore the component of $\Omega^1_D(\mathcal{A})$ along K_7 is trivial: K_7 generates the zero one-form.

The spectral action $\mathrm{Tr}\, f(D_A^2/\Lambda^2)$ expanded to second order in a gauge potential A produces the gauge kinetic term through the Seeley–DeWitt coefficient a_4. This coefficient contains terms of the form $\mathrm{Tr}(F^a_{\mu\nu} F^{a\,\mu\nu})$, where $F^a_{\mu\nu}$ is the field strength of the gauge field $A^a_\mu = \sum_i a_i [D, b_i]_\mu$ associated to the generator b of the gauge symmetry. Since $[D_K, K_7] = 0$, we have $A_7 = 0$, $F_7 = 0$, and no kinetic term appears. The $U(1)_7$ gauge boson is absent from the spectral action. □

Corollary 7 (Non-perturbative extension)**.** *The commutant obstruction holds at all orders in perturbation theory and extends non-perturbatively.*

Proof. If $[K_7, D_K] = 0$, then $[K_7, p(D_K)] = 0$ for every polynomial p (by induction: $[K_7, D_K^{n+1}] = [K_7, D_K^n] D_K + D_K^n [K_7, D_K] = 0$). More generally, $[K_7, f(D_K)] = 0$ for every measurable function f: since K_7 commutes with D_K, it commutes with each spectral projection E_λ of the spectral measure $\{E_\lambda\}$ (by the double commutant theorem applied to the von Neumann algebra generated by D_K), and hence with $f(D_K) = \int f(\lambda)\, dE_\lambda$.

The one-loop counterterms of the spectral action have the form $\Sigma(D_K) = c_0 + c_2 D_K^2 + c_4 D_K^4 + \cdots$ van Nuland and van Suijlekom [2022], so the effective Dirac operator $D_{\text{eff}} = D_K + \Sigma(D_K)$ satisfies

$$[iK_7, D_{\text{eff}}] = [iK_7, D_K] + [iK_7, \Sigma(D_K)] = 0 + 0 = 0. \tag{15}$$

The same argument applies at every loop order, and the spectral theorem extends it to arbitrary (non-polynomial) functionals of D_K. The one-form obstruction $A_7 = a[D_{\text{eff}}, K_7] = 0$ therefore holds non-perturbatively.

Numerically, the propagation has been verified explicitly:

$$\begin{aligned} \|[iK_7, D_K^2]\|_F / \|D_K^2\|_F &= 1.3 \times 10^{-17}, \\ \|[iK_7, D_K^4]\|_F / \|D_K^4\|_F &= 3.0 \times 10^{-17}, \\ \|[iK_7, e^{-D_K^2/\Lambda^2}]\|_F / \|e^{-D_K^2/\Lambda^2}\|_F &= 3.1 \times 10^{-16}, \end{aligned} \tag{16}$$

where the last line uses $\Lambda = 3\, M_{\text{KK}}$ Project computation [2026f]. All ratios are at machine epsilon. □

4.2 Proof II: Categorical (inner vs. outer)

This proof establishes the impossibility from the structure theory of automorphisms, *without* using the commutant property $[K_7, D_K] = 0$. It applies regardless of whether K_7 commutes with the Dirac operator.

Proof II. The argument has four steps.

Step 1: K_7 generates a diffeomorphism. The Kosmann derivative K_7 is the canonical lift (9) of the Killing vector field $X_7 = \lambda_7$ to the spinor bundle $\Sigma(\text{SU}(3))$. The flow $\phi_t = \exp(tX_7)$ is a one-parameter group of isometries of $(\text{SU}(3), g_\tau)$, and K_7 generates its lift to the spin structure. In particular, K_7 generates a one-parameter group of diffeomorphisms acting on sections of $\Sigma(\text{SU}(3))$.

Step 2: Diffeomorphisms are automorphisms of $C^\infty(\text{SU}(3))$. A diffeomorphism $\phi : \text{SU}(3) \to \text{SU}(3)$ induces an algebra automorphism $\alpha_\phi : C^\infty(\text{SU}(3)) \to C^\infty(\text{SU}(3))$ by pullback: $\alpha_\phi(f) = f \circ \phi$. The Gelfand–Naimark theorem identifies $\text{Aut}(C^\infty(M)) \cong \text{Diff}(M)$ for compact manifolds M Connes [1994].

Step 3: For commutative algebras, all automorphisms are outer. For a commutative unital $*$-algebra $\mathcal{A}$, conjugation by any unitary $u \in U(\mathcal{A})$ is trivial: $uau^* = uu^*a = a$ for all $a \in \mathcal{A}$. Therefore $\text{Inn}(\mathcal{A}) = \{\text{id}\}$, and every nontrivial automorphism is outer:

$$\mathcal{A} \text{ commutative} \quad \Longrightarrow \quad \text{Out}(\mathcal{A}) = \text{Aut}(\mathcal{A}), \quad \text{Inn}(\mathcal{A}) = \{\text{id}\}. \tag{17}$$

In particular, the automorphism α_{ϕ_t} induced by the flow of K_7 is outer for $C^\infty(\text{SU}(3))$.

Step 4: Gauge fields come from $\text{Inn}(\mathcal{A}_F)$, not from diffeomorphisms. In the almost-commutative spectral triple of Definition 2, $\mathcal{A} = C^\infty(\text{SU}(3)) \otimes \mathcal{A}_F$. The gauge group is $\mathcal{G} = \text{Inn}(\mathcal{A}) = \text{Inn}(\mathcal{A}_F) \cong U(1)_Y \times \text{SU}(2)_L \times \text{SU}(3)_c$. Gauge connections are inner fluctuations $A = \sum_i a_i[D, b_i]$ with $a_i, b_i \in \mathcal{A}$, and the gauge fields that participate in the Anderson–Higgs mechanism are precisely those arising from $\text{Inn}(\mathcal{A}_F)$.

The generator $K_7 \otimes \mathbf{1}_F$ acts as a diffeomorphism of SU(3) tensored with the identity on $\mathcal{A}_F$. It is an outer automorphism of the first factor and the identity on the second. It is therefore outer for the product algebra $\mathcal{A}$. Since gauge fields arise exclusively from inner automorphisms Connes [1996], K_7 cannot produce a gauge field. □

Remark 8 (Independence of the two proofs). Proof I uses $[K_7, D_K] = 0$ and is independent of the algebra structure beyond the existence of $\Omega^1_D(\mathcal{A})$. Proof II uses only the categorical position of K_7 as a diffeomorphism generator and the commutativity of $C^\infty(\mathrm{SU}(3))$ – it does *not* invoke $[K_7, D_K] = 0$ at any step. Proof II would hold even for a hypothetical Killing field whose Kosmann derivative did *not* commute with D_K. The two arguments are therefore structurally distinct.

Corollary 9 (Anderson–Higgs impossibility)**.** *The $U(1)_7$ symmetry of $D_K(\tau)$ on Jensen-deformed* SU(3) *admits no Anderson–Higgs mechanism within the almost-commutative spectral triple of Definition 2. Specifically:*

1. *There exists no gauge boson A_7^μ in the spectral action for $U(1)_7$.*
2. *The Goldstone mode of $U(1)_7$ breaking cannot be absorbed.*
3. *No mass is generated for the $U(1)_7$ sector through the Higgs mechanism.*

Remark 10 (D_{phys} and the fluctuated commutant). The ungaugeability theorem (Theorem 6) applies to the *unfluctuated* Dirac operator D_K. When inner fluctuations from $\mathcal{A}_F$ are turned on, the fluctuated operator

$$D_{\text{phys}} = D_K + A + \varepsilon' J A J^{-1}, \qquad A \in \Omega^1_D(\mathcal{A}),$$

may have a different commutant. Concretely, at gap-edge amplitude $\phi = 0.133$:

$$\|[iK_7, D_{\text{phys}}]\|_F / \|D_{\text{phys}}\|_F = 0.117 \quad \textit{Projectcomputation}\ [2026f].$$

This 11.7% breaking of $[iK_7, D_{\text{phys}}] = 0$ shows that K_7 does *not* lie in the commutant of D_{phys}, and one might ask whether this opens a route to gauging $U(1)_7$.

It does not. The breaking arises from inner fluctuations of $\mathcal{A}_F$ – gauge fields for $\mathrm{SU}(2)_L \times U(1)_Y$, not for $U(1)_7$. Gaugeability in NCG requires $K_7 \in \mathrm{Inn}(\mathcal{A})$, which is a categorical property of K_7's position in the automorphism group (Proof II). Whether K_7 commutes with D_{phys} or not is irrelevant: K_7 generates a diffeomorphism of SU(3), which is outer for $C^\infty(\mathrm{SU}(3))$, and no fluctuation of D can promote an outer automorphism to an inner one. The commutant breaking by D_{phys} means that $U(1)_7$ ceases to be an *exact symmetry* of the fluctuated operator, but it was never a *gauge symmetry* to begin with.

5 Consequences

5.1 The Goldstone boson persists

In condensed matter and particle physics, when a continuous symmetry is spontaneously broken, Goldstone's theorem guarantees the existence of a massless boson. The Anderson–Higgs mechanism provides the sole known escape: if the broken symmetry is gauged, the Goldstone boson is absorbed as the longitudinal polarization of the massive gauge boson.

For $U(1)_7$ on Jensen-deformed SU(3), the BCS-type condensate formed from eigenmodes of D_K carries K_7 charge $\pm 1/2$ and spontaneously breaks $U(1)_7$ Project computation [2026d]. The Goldstone mode associated to this breaking has frequency $\omega_G \approx 0.070\, M_{\text{KK}}$ Project computation [2026e]. By Theorem 6, no gauge boson exists to absorb it. The Goldstone mode is permanent and ungapped within the NCG inner fluctuation framework.

5.2 Implications for Kaluza–Klein spectral triples

In a Kaluza–Klein geometry $M^4 \times K$ where K is a compact internal manifold, the isometry group $\mathrm{Iso}(K)$ acts on the spectrum of the internal Dirac operator through Kosmann derivatives. The Jensen deformation provides a smooth family of metrics on $K = \mathrm{SU}(3)$ that breaks $\mathrm{Iso}(K)$ from $\mathrm{SU}(3) \times \mathrm{SU}(3)$ to $\mathrm{SU}(3)_L \times U(1)_7$.

Theorem 6 shows that this residual $U(1)_7$ is *structurally different* from the gauge symmetries $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times U(1)_Y$ that arise from the algebra $\mathcal{A}_F$. The isometry symmetry and the gauge symmetry belong to categorically distinct classes of automorphisms – outer and inner, respectively – and no mechanism within the spectral action framework can convert one into the other.

This has a broader implication: in any KK spectral triple where an isometry of K commutes with D_K, that isometry is automatically ungaugeable. The commutant condition $[K_X, D_K] = 0$ is both the signature of an isometry (via the Kosmann–Lichnerowicz theorem) and the obstruction to gauging (via the trivial one-form argument). These two facts are the same mathematical statement viewed from two perspectives.

5.3 Spectral action and the inner/outer dichotomy

The spectral action principle Chamseddine and Connes [1996] asserts that the entire bosonic Lagrangian is encoded in $\mathrm{Tr}\, f(D^2/\Lambda^2)$. Theorem 6 reveals a structural feature of this principle: the spectral action *automatically* excludes gauge fields for symmetries generated by commutant elements. This is not a defect of the formalism but a reflection of its internal consistency: the spectral action generates dynamics only for the degrees of freedom that appear as inner fluctuations, and the commutant directions are, by definition, the directions in which D does not fluctuate.

The inner/outer dichotomy thus acquires a dynamical meaning through the spectral action: inner automorphisms produce propagating gauge fields with kinetic terms in a_4; outer automorphisms (including Kosmann symmetries in the commutant) produce global symmetries with conservation laws but no propagating bosons.

6 Relation to Prior Work

6.1 Connes' inner/outer distinction

The classification of automorphisms into inner (gauge) and outer (diffeomorphisms) is a foundational element of Connes' program, established in Connes [1994, 1996] and reviewed in Connes [2021], Connes and Marcolli [2008]. The gauge group derivation for the Standard Model – showing that $\mathrm{Inn}(\mathcal{A}_F) \cong U(1)_Y \times \mathrm{SU}(2)_L \times \mathrm{SU}(3)_c$ for $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ – is one of the central results of NCG applied to particle physics Chamseddine et al. [2007]. Our work does not extend or modify this framework; rather, we apply it to a specific manifold and symmetry to derive a concrete impossibility result.

6.2 Kosmann's construction

The Lie derivative of spinors was defined by Kosmann in 1971 Kosmann [1971] and subsequently studied by Lichnerowicz Lichnerowicz [1963] and others. The key properties – the Leibniz rule with respect to Clifford multiplication, commutation with the Dirac operator for Killing fields, and the structure (9) as a first-order differential operator on sections of the spinor bundle – are classical. Recent work by Baptista Baptista [2024, 2025] has placed the Kosmann derivative in the context of Kaluza–Klein theory, showing how it governs the coupling of massive gauge fields to internal spinor modes. We note an additional structural distinction: the Kosmann derivative is a *first-order differential operator* (involving ∇^{Σ}_{X} and $\nabla_a X_b$), while inner fluctuations $a[D, b]$ are *zeroth-order* (multiplication operators, since $[D, f] = -i\gamma^{\mu}\partial_{\mu} f$ is zeroth-order in derivatives when $f \in C^{\infty}(M)$). This difference in operator order provides an independent perspective on why Kosmann derivatives cannot arise as inner fluctuations, complementing the categorical argument of Proof II.

6.3 Inner fluctuations without the first-order condition

Chamseddine, Connes, and van Suijlekom Chamseddine et al. [2013] extended the inner fluctuation framework to spectral triples that violate the first-order condition $[[D, a], b^{\circ}] = 0$. In this generalized setting, quadratic terms $\sum c_{ij}[D, a_i][D, a_j]$ appear in the fluctuated Dirac operator. The Dirac operator $D_K(\tau)$ on Jensen-deformed SU(3) violates the first-order condition at the maximal Clifford value of 4.000 for $(\mathbb{H}, \mathbb{H})$ pairs Project computation [2026a]. One might ask whether the quadratic fluctuations could circumvent the commutant obstruction.

They cannot: the quadratic term involves products $[D_K, a_i][D_K, a_j]$ with $a_i, a_j \in \mathcal{A}$. Since $[D_K, K_7] = 0$, any product involving K_7 as one of the arguments vanishes. The commutant obstruction persists even in the generalized fluctuation framework.

6.4 What is new

The inner/outer distinction is known. The Kosmann derivative is known. The spectral action framework is known. For the commutative algebra $C^{\infty}(\mathrm{SU}(3))$, the ungaugeability of any diffeomorphism follows from general principles ($\mathrm{Inn} = \{\mathrm{id}\}$). What is new in this paper is:

1. An *explicit worked example* demonstrating ungaugeability of a Kosmann symmetry in a concrete Kaluza–Klein setting – the Jensen-deformed SU(3) with an almost-commutative spectral triple – with computational verification of $[iK_7, D_K(\tau)] = 0$ across the full deformation family.

2. The identification that the commutant property and the spectral theorem together close the obstruction *non-perturbatively* (Corollary 7), not merely at tree level.

3. The *physical consequence*: the impossibility of the Anderson–Higgs mechanism for $U(1)_7$, the persistence of the Goldstone mode, and the resulting constraint on mass generation in Kaluza–Klein spectral triples on compact Lie groups.

7 Conclusion

We have proven that the $U(1)_7$ symmetry of the Dirac operator on Jensen-deformed SU(3), generated by the Kosmann derivative K_7, cannot be gauged within the framework of Connes' noncommutative geometry. Two structurally distinct arguments establish this: the commutant obstruction (Proof I, with its non-perturbative extension via the spectral theorem) and the categorical inner/outer classification (Proof II, which is independent of the commutant property). The Anderson–Higgs impossibility for $U(1)_7$ follows as a direct corollary.

The result rests on two complementary pillars. The commutant property $[K_7, D_K] = 0$ simultaneously certifies K_7 as a symmetry generator (Kosmann–Lichnerowicz theorem) and obstructs its gauging (trivial one-form in $\Omega^1_D(\mathcal{A})$). Independently, the categorical position of K_7 as a diffeomorphism generator – outer for the commutative algebra $C^\infty(\mathrm{SU}(3))$ – excludes it from the gauge group $\mathrm{Inn}(\mathcal{A}_F)$ regardless of its commutant status.

The result is a structural wall in the space of mass-generation mechanisms for KK spectral triples. Any symmetry arising from an isometry of the internal manifold that commutes with the internal Dirac operator is permanently excluded from the gauge sector. The spectral action, by its dependence on the spectrum of D alone, inherits and enforces this exclusion automatically.

Several directions for further investigation present themselves. First, the classification of *all* commutant elements of $D_K(\tau)$ – not just K_7 – would determine the full space of ungaugeable symmetries for each value of τ. Second, the fluctuated operator D_{phys} breaks $[iK_7, D_{\mathrm{phys}}] \neq 0$ (Remark 10); while this does not enable gauging (by Proof II), it raises the question of whether the broken $U(1)_7$ has observable consequences in the low-energy effective theory. Third, the interplay between ungaugeability and the persistence of the BCS Goldstone mode may constrain the physical spectrum of Kaluza–Klein compactifications on group manifolds.

References

J. M. Baptista. Internal symmetries in Kaluza–Klein models. *arXiv preprint*, 2024. arXiv:2403.XXXXX.

J. M. Baptista. Chiral interactions of fermions and massive gauge fields in Kaluza–Klein models. *arXiv preprint*, 2025. arXiv:2501.XXXXX.

Ali H. Chamseddine and Alain Connes. The spectral action principle. *Communications in Mathematical Physics*, 186(3):731–750, 1996. doi: 10.1007/s002200050126. arXiv:hep-th/9606001.

Ali H. Chamseddine, Alain Connes, and Matilde Marcolli. Gravity and the standard model with neutrino mixing. *Advances in Theoretical and Mathematical Physics*, 11(6):991–1089, 2007. arXiv:hep-th/0610241.

Ali H. Chamseddine, Alain Connes, and Walter D. van Suijlekom. Inner fluctuations in noncommutative geometry without the first order condition. *Journal of Geometry and Physics*, 73:222–234, 2013. doi: 10.1016/j.geomphys.2013.06.006. arXiv:1304.7583.

Alain Connes. *Noncommutative Geometry*. Academic Press, San Diego, 1994.

Alain Connes. Noncommutative geometry and reality. *Journal of Mathematical Physics*, 36 (11):6194–6231, 1995. doi: 10.1063/1.531241.

Alain Connes. Gravity coupled with matter and the foundation of non-commutative geometry. *Communications in Mathematical Physics*, 182(1):155–176, 1996. doi: 10.1007/BF02506388. arXiv:hep-th/9603053.

Alain Connes. Noncommutative geometry, the spectral standpoint. In *New Spaces in Mathematics*, pages 23–58. Cambridge University Press, 2021. arXiv:1910.10407.

Alain Connes and Matilde Marcolli. *Noncommutative Geometry, Quantum Fields and Motives*, volume 55 of *Colloquium Publications*. American Mathematical Society, 2008.

Gary R. Jensen. Einstein metrics on principal fibre bundles. *Journal of Differential Geometry*, 8(4):599–614, 1973. doi: 10.4310/jdg/1214431962.

Yvette Kosmann. Dérivées de Lie des spineurs. *Annali di Matematica Pura ed Applicata*, 91:59–81, 1971. doi: 10.1007/BF02428822.

André Lichnerowicz. *Spineurs harmoniques*, volume 257. 1963. See also: Géométrie des groupes de transformations, Dunod, Paris, 1958.

Project computation. Order-one condition violation for d_k on SU(3), 2026a. Sessions 9–10: $[[D_K, a], b^\circ] = 4.000$ for $(\mathbb{H}, \mathbb{H})$ pairs.

Project computation. Volume-preserving Jensen deformation of SU(3), 2026b. Script: `tier1_dirac_spectrum.py`.

Project computation. Verification of $[ik_7, d_k(\tau)] = 0$ on Jensen-deformed SU(3), 2026c. Script: `s34a_dphys_fold.py`, 79,968 matrix pairs, 9 τ-values.

Project computation. k_7 charge eigenvalues and Cooper pair charge in BCS condensate, 2026d. Script: `s35_pfaffian_j_corrected.py`, 34 τ-values.

Project computation. Leggett mode frequency and mass problem characterization, 2026e. Session 50 cross-domain investigation.

Project computation. GAUGE-U1K7-51: $\mathrm{U}(1)_7$ gauging via inner fluctuations – three-level impossibility, 2026f. Script: `s51_gauge_u1k7.py`, data: `s51_gauge_u1k7.npz`.

Teun D. H. van Nuland and Walter D. van Suijlekom. One-loop corrections to the spectral action. *Journal of High Energy Physics*, 2022:078, 2022. doi: 10.1007/JHEP05(2022)078. arXiv:2107.08485.